Homework 5.1-5.2 #1

Student Name: ____________________________

Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **9**

Instructor Name: **Mamani Velasco, Ruben**

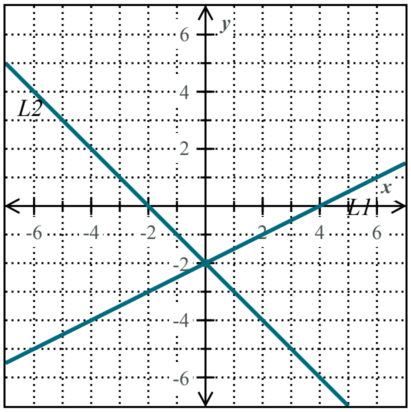
Question 1 of 9

For each system of linear equations shown below, classify the system as "consistent dependent," "consistent independent," or "inconsistent." Then, choose the best description of its solution. If the system has exactly one solution, give its solution.

System A

L1: $y = \frac{1}{2}x - 2$

L2: $y = -x - 2$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

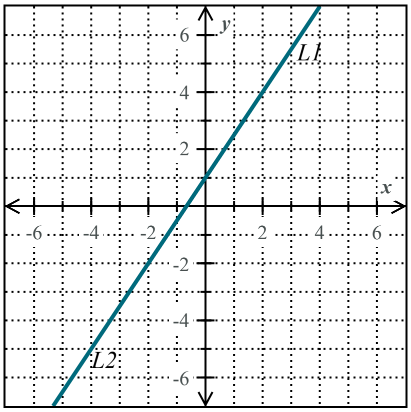
- a unique solution: - no solution - infinitely many solutions

Solution: $(\_\_\_ , \_\_\_)$

System B

L1: $y = \frac{3}{2}x + 1$

L2: $-3x + 2y = 2$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

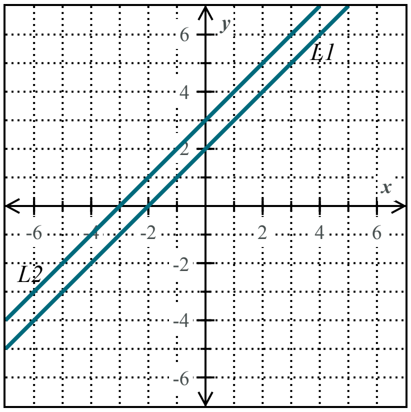
- a unique solution: - no solution - infinitely many solutions

Solution: $(\_\_\_ , \_\_\_)$

System C

L1: $y = x + 2$

L2: $y = x + 3$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

- a unique solution: - no solution - infinitely many solutions

Solution: $(\_\_\_ , \_\_\_)$

Question 2 of 9

Solve the following system of equations.

$$-3x - 5y = -9$$
$$5x - 2y = -16$$

Question 3 of 9

Solve the following system of equations.

$$0.8x - 0.6y = 12$$
$$0.6x - 0.2y = 7$$

Solve the following system of equations.

$$\begin{cases} 0.8x - 0.6y = 12 \\ 0.6x - 0.2y = 7 \end{cases}$$

$x =$ ______

$y =$ ______

Question 4 of 9

Two systems of equations are given below.
For each system, choose the best description of its solution.
If applicable, give the solution.

(a) Solve the following system of equations.

$$\begin{cases} -x+4y=8 \\ x-4y=8 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\square, \square)$
- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \square$

(b) Solve the following system of equations.

$$\begin{cases} -x-2y-4=0 \\ x-2y=4 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\square, \square)$
- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \square$

(a) Solve the following system of equations.

$$\begin{cases} -x+4y=8 \\ x-4y=8 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\_\_\_, \_\_\_)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \_\_\_$

(b) Solve the following system of equations.

$$\begin{cases} -x-2y-4=0 \\ x-2y=4 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\_\_\_, \_\_\_)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \_\_\_$

Question 5 of 9

A party rental company has chairs and tables for rent. The total cost to rent 3 chairs and 5 tables is $\$45$. The total cost to rent 6 chairs and 2 tables is $\$24$. What is the cost to rent each chair and each table?

Question 6 of 9

An auto shop has two mechanics. They charge a combined rate of $\$160$ per hour. On their most recent car, the first mechanic worked for 20 hours, and the second mechanic worked for 15 hours. Together they charged a total of $\$2975$.

What is the rate charged per hour by each mechanic?

First mechanic: $\$$ ______ per hour

Second mechanic: $\$$ ______ per hour

Question 7 of 9

Two trains leave towns 1335 kilometers apart at the same time and travel toward each other. One train travels $17 \frac{\text{km}}{\text{h}}$ faster than the other.

If they meet in 5 hours, what is the rate of each train?

Rate of the faster train: ______ $\frac{\text{km}}{\text{h}}$

Rate of the slower train: ______ $\frac{\text{km}}{\text{h}}$

Question 8 of 9

Solve the following system.

$$\begin{cases} z = 5 \\ 2y + 3z = 3 \\ 3x - 5y - 4z = -2 \end{cases}$$

Question 9 of 9

Solve the system.

$$\begin{aligned} 2x - y + 2z &= 7 \\ -2x + 2y - z &= -6 \\ -4x + 3y - 2z &= -9 \end{aligned}$$

$x =$

$y =$

$z =$

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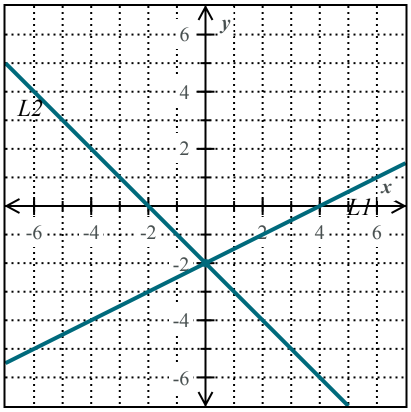
Question 1 of 9

System A

L1: $y = \dfrac{1}{2}x - 2$

L2: $y = -x - 2$



This system of equations is:

- consistent independent

This means the system has:

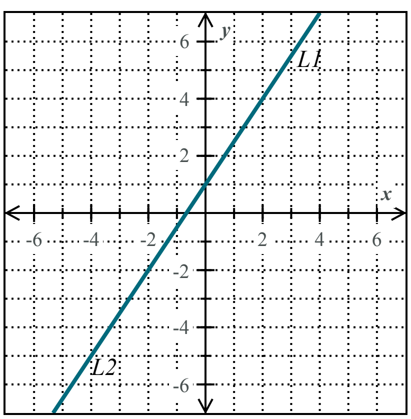
- a unique solution:

Solution: $(0, \ -2)$

System B

L1: $y = \dfrac{3}{2}x + 1$

L2: $-3x + 2y = 2$



This system of equations is:

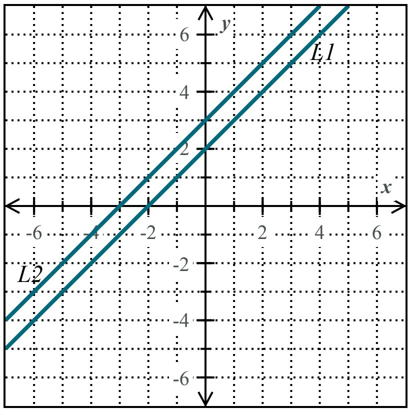
- consistent dependent

This means the system has:

- infinitely many solutions

System C

L1: $y = x + 2$
L2: $y = x + 3$



This system of equations is:

- inconsistent

This means the system has:

- no solution

Question 2 of 9

$x = -2$

$y = 3$

Question 3 of 9

$x = 9$
$y = -8$

Question 4 of 9

(a) ◉ The system has no solution.

○ The system has a unique solution.

$(x, y) = (\square, \square)$

○ The system has infinitely many solutions. They must satisfy the following equation.

$y = \square$

(b) ○ The system has no solution.

◉ The system has a unique solution.

$(x, y) = (0, -2)$

○ The system has infinitely many solutions. They must satisfy the following equation.

$y = \square$

Question 5 of 9

Cost to rent each chair: $\$1.25$

Cost to rent each table: $\$8.25$

Question 6 of 9

First mechanic: $\$115$ per hour

Second mechanic: $\$45$ per hour

Question 7 of 9

If they meet in 5 hours, what is the rate of each train?

Rate of the faster train: $142 \ \frac{\text{km}}{\text{h}}$

Rate of the slower train: $125 \ \frac{\text{km}}{\text{h}}$

Question 8 of 9

$x = -4$

$y = -6$

$z = 5$

Question 9 of 9

$x = -2$

$y = -3$

$z = 4$